Q P Code:-25901395

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU- UGP (HONOURS) REGULAR EXAMINATION March 2026

FOURTH SEMESTER

Skill Enhancement Course (SEC) - MG4SECMAT200
DOCUMENT PREPARATION USING LaTeX

(2024 ADMISSION ONWARDS)

Scheme For Valuation

Duration: 1.5 Hours Maximum Marks: 50

Part A

(Short Answer Type Questions)
Answer any 5 questions
Each question carries 2 marks

Question 1 What command is used to start a new item in a list? **[U] / [CO1]**

- `\item` ...2 Marks
- Any example like

```
\begin{enumerate/itemize/description}
 \item .....
\end{enumerate/itemize/description}
```

...2 Marks

Total Marks should not Exceed 2

Question 2 What type of file is used as input for LaTeX? **[U] / [CO1]**

- The input to LaTeX is simply a text file. ...2 Marks

Question 3 Write the LaTeX command to type the symbols \,{ and } **[S] / [CO2]**

- `\textbackslash` for \ ...1 Mark
- `\{` for { ...1 Mark
- `\}` for } ...1 Mark

Total Marks should not Exceed 2

Question 4 Which pair of commands do you use in the document body to activate a user-defined environment? **[U] / [CO3]**

- `\newenvironment{name}{begin text}{end text}`

. . . 2 Marks

- Any example like

  `\newenvironment{emphit}{\begin{itemize}}{\end{itemize}}`

. . . 2 Marks

Total Marks should not Exceed 2

Question 5 Which two environments allow LaTeX to move content to the top or bottom of a page to avoid gaps? **[U] / [CO2]**

- The two primary environments in LaTeX that allow content to move to the top or bottom of a page to avoid gaps are `figure` and `table`. . . . 2 Marks

Question 6 What is the purpose of the overlay environment in slides? **[S] / [CO5]**

- The overlay environment is used for an overlay-a slide that is meant to be put on top of another slide. . . . 2 Marks

Question 7 Write the output of the following LaTeX code

```
\begin{thebibliography}{99}
\bibitem{al} Ponnusamy, S., Herb Silverman.
\textbf{Complex variables with applications.}
Springer Science \& Business Media, 2007.
\end{thebibliography}
```

[C] / [CO4]

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Bibliography

[1] Ponnusamy, S., Herb Silverman. **Complex variables with applications.** Springer Science & Business Media, 2007.

. . . 2 Marks

N.B:- Neglect minor mistakes

Question 8 What is the function of the `\nocite` command? **[[S] / [CO4]**

- The command `\nocite{key_list}` in LaTeX is used to include references in the bibliography without actually citing them in the document text. . . . 2 Marks

Total Marks should not Exceed 2

Question 1 **[K] / [C01]**

- Correct Definition . . . 2 Marks

Total Marks should not Exceed 2

Part B
(Short Essay Questions)
Answer any 5 questions
Each question carries 6 marks

Question 9 Write a short note on math styles in LaTeX? What declarations are used to force each math style? **[A] / [CO2]**

- LaTeX provides the following commands for changing type style in math mode:

LaTeX Code	Output
`\[\mathit{italic}\]`	*italic*
`\[\mathrm{roman}`	roman
`\[\mathbf{bold}`	**bold**
`\[\mathsf{sansserif}`	sansserif
`\[\mathtt{typewriter}`	typewriter
`\[\mathcal{UPPERCASE}\]`	$\mathcal{PPERCASE}$

. . . 5 Marks

N.B:- Neglect minor mistakes

Question 10 With proper examples write the purpose of the following LaTeX commands

(a) `\upshape`

(b) `\itshape`

(c) `\slshape`

[S] / [CO3]

- `\upshape` used for declaration to Upright shape
  For example `\upshape Text Style` $\longrightarrow$ Text Style . . . 2 Marks
- `\itshape` used for declaration to Italic shape
  For example `\itshape Text Style` $\longrightarrow$ *Text Style* . . . 2 Marks
- `\slshape` used for declaration to Slanted shape
  For example `\slshape Text Style` $\longrightarrow$ *Text Style* . . . 2 Marks

Total Marks should not Exceed 5

Question 11 Describe how LaTeX places floating objects in a document. **[U] / [CO2]**

- In LaTeX, floating objects such as figures and tables are placed using the `figure` and `table` environments. These are called *floats* because their exact position is not fixed; instead, LaTeX determines their placement to achieve a well-balanced page layout. . . . 5 Marks

Question 12 Write the output of the following LaTeX code

```
\newtheorem{thm}{Theorem}
\begin{thm}[Euclid]
The sum of the angles of a triangle is 180 degree
\end{thm}
```

[U] / [CO3]

- **Theorem 1 (Euclid)** *The sum of the angles of a triangle is 180 degree*

. . . 5 Marks

Question 13 Explain with an example, how can we give `\label` command in eqarray environment **[A] / [CO3]**

- In LaTeX, the `\label` command is used inside the equation environment (including eqnarray) to assign a reference name to an equation so that it can be cited later using `\ref`. . . . 2 Marks
- In the following example ,

```
We consider the following system of equations:
    \begin{eqnarray}
    a + b &=& c \label{eq:first} \\
    x + y &=& z \label{eq:second}
    \end{eqnarray}
Equation~\ref{eq:first} represents the first relation, while
Equation~\ref{eq:second} represents the second.
    \end{itemize}
```

Here

(a) `\label{eq:first}` assigns a label to the first equation.

(b) `\label{eq:second}` assigns a label to the second equation.

(c) `\ref{eq:first}` and `\ref{eq:second}` are used to refer to those equations in the text.

. . . 5 Marks

Hence the output will be

We consider the following system of equations:

$$a + b = c \quad (1)$$
$$x + y = z \quad (2)$$

Equation 1 represents the first relation, while Equation 2 represents the second.

. . . 2 Marks

Total Marks should not Exceed 5

Question 14 Explain the use of the following commands in book class

(a) `\openany`

(b) `\openright`

(c) `\backmatter`

[S] / [CO2]

- `\openany`
  This command allows chapters to begin on any page (either odd or even). By default, the book class often starts chapters on right-hand (odd-numbered) pages, which may leave blank pages. Using `\openany` avoids unnecessary blank pages. . . . 3 Marks
- `\openright`
  This command forces each chapter to start on a right-hand (odd-numbered) page. If the previous chapter ends on an odd page, LaTeX inserts a blank page. . . . 3 Marks
- `\backmatter`
  This command is used to begin the back matter of the book. Typically used before sections like:
  (a) Bibliography
  (b) Index
  (c) Appendix etc.

. . . 3 Marks

Total Marks should not Exceed 5

Part C
(Essay Questions)

Answer any 3 questions
Each question carries 10 marks

Question 15 (a) Give the LaTeX commands for following mathematical formulas

i $\sqrt[3]{x^3}$

ii $\frac{y+z/2}{y^2+1}$

iii $\sum_{i=1}^{n} x_i = \int_0^{\pi} f$

[S] / [CO2]

- `\(\sqrt[3]{x^3}\)` ...4 Marks
- `\(\frac{y + z / 2}{y^2 + 1}\)` ...4 Marks
- `\(\sum_{i = 1}^{n} x_{i} = \int_{0}^{\pi} f\)` ...4 Marks

Total Marks should not Exceed 10

Question 16 (a) Write LaTeX file to produce the following output (without outer frame)

Thus x, y and z satisfy the equations

$$x + y - z = 1$$

$$x - y + z = 1$$

and by hypothesis

$$x + y + z = 1 \quad (1)$$

] **[An] / [CO2]**

- ```
  Thus \( x,y\) and \(z\) satisfy the equations
  \begin{equation*}
   x + y - z = 1
  \end{equation*}
  \begin{equation*}
   x - y + z = 1
  \end{equation*}
  and by hypothesis
  \begin{equation}
   \[x + y + z = 1
  \end{equation}
  ```

...6 Marks
  ```

(b) Write the output of the following LaTeX code

```
\begin{equarray}
(a+b+c+d+e+f)^2 & = a^2+b^2+c^2+d^2+e^2+f^2 \\
&+2ab+2ac+2ad+2ae+2af \\
&+2bc+2bd+2be+2bf \\
&+2cd+2ce+2cf \\
&+2de+2df \\
& +2ef
\end{equarray}
```

- $$\begin{aligned}(a+b+c+d+e+f)^2 = a^2+b^2+c^2+d^2+e^2+f^2 \\ +2ab+2ac+2ad+2ae+2af \\ +2bc+2bd+2be+2bf \\ +2cd+2ce+2cf \\ +2de+2df \\ +2ef\end{aligned}$$

. . . 6 Marks

Total Marks should not Exceed 10

Question 17 (a) Explain how `\newcommand` helps in LaTeX.

- The simplest repeated structure occurs when the same text or mathematical notation appears in different places in a document. The `\newcommand` declaration is used to define a new command for such repeated text.

. . . 4 Marks

- The syntax is

```
\newcommand{\commandname}{text}
```

. . . 1 Mark

(b) Write the LaTeX code using `\newcommand` to create a shortcut `\smat` for the course name "Skill Enhancement Course in Mathematics".

```
\newcommand{\smat}{Skill Enhancement Course in Mathematics}
```

. . . 5 Marks

(c) Write the output of the following LaTeX code

```
\newcommand{\vect}[2]{(#1_1,#1_2,\dotsc,#1_2)}
$\vect{x}{n},\vect{y}{m},\vect{z}{k}$
```

- $(x_1, x_2, \ldots, x_2), (y_1, y_2, \ldots, y_2), (z_1, z_2, \ldots, z_2)$ ... 5 Marks

Total Marks should not Exceed 10

Question 18 Write LaTeX code to produce an equation with a label and refer to it later in the text. Show the expected output. **[S]/[CO3]**

- Any example like,

```
Here is an important equation:

\begin{equation}
E = mc^2
\label{eq:einstein}
\end{equation}

As shown in equation~(\ref{eq:einstein}), energy is equal
to mass times the square of the speed of light.
```

... 6 Marks

- Its output will be,

Here is an important equation:

$$E = mc^2 \tag{1}$$

As shown in equation (1), energy is equal to mass times the square of the speed of light.

... 6 Marks

Total Marks should not Exceed 10

N.B: Suitable stage marks may be fixed by the examiner for alternate correct answers.